Experiment no 5

Stemming in English using NLTK:

Program:

```
from nltk.stem import PorterStemmer
from nltk.stem import LancasterStemmer
from nltk.stem import SnowballStemmer

# Create stemmer objects
porter = PorterStemmer()
lancaster = LancasterStemmer()
snowball = SnowballStemmer("english")

# Input words
words = [
    "caresses",
    "ponies",
    "ties",
    "caress",
    "cats",
    "feed",
    "agreed",
    "plastered",
    "bled",
    "motoring",
    "relational",
    "conditional",
    "rational",
    "digitizer",
    "vietnamization",
    "predication",
    "operator",
    "feudalism",
    "decisiveness",
    "hopefulness",
    "callousness",
    "formalities",
    "sensitivities",
    "triplicate",
```

```
    "formative",
    "formalize",
    "electrical",
    "hopeful",
    "goodness",
    "electrical",
]

# Display output
print("=" * 75)
print("            STEMMING USING NLTK")
print("=" * 75)

print(f"{'WORD':<20}{'PORTER':<20}{'LANCASTER':<20}{'SNOWBALL':<20}")
print("-" * 75)

for word in words:
    porter_result = porter.stem(word)
    lancaster_result = lancaster.stem(word)
    snowball_result = snowball.stem(word)

    print(
        f"{word:<20}"
        f"{porter_result:<20}"
        f"{lancaster_result:<20}"
        f"{snowball_result:<20}"
    )

print("=" * 75)
```

Output

```
================================================================================
             STEMMING USING NLTK
================================================================================
WORD                PORTER              LANCASTER           SNOWBALL
--------------------------------------------------------------------------------
caresses            caress              caress              caress
ponies              poni                pony                poni
ties                tie                 tie                 tie
caress              caress              caress              caress
cats                cat                 cat                 cat
feed                feed                fee                 feed
agreed              agre                agree               agre
plastered           plaster             plast               plaster
bled                bled                bled                bled
motoring            motor               mot                 motor
relational          relat               rel                 relat
conditional         condit              condit              condit
rational            ration              rat                 ration
digitizer           digit               digit               digit
vietnamization      vietnam             vietnam             vietnam
predication         predic              pred                predic
operator            oper                op                  oper
feudalism           feudal              feud                feudal
decisiveness        decis               decid               decis
hopefulness         hope                hop                 hope
callousness         callous             cal                 callous
formalities         formal              form                formal
sensitivities       sensit              sensit              sensit
triplicate          triplic             triply              triplic
formative           form                form                format
formalize           formal              form                formal
electrical          electr              elect               electr
hopeful             hope                hop                 hope
goodness            good                good                good
electrical          electr              elect               electr
================================================================================
```

Stemming in English using Spacy and Stanza: Not Available

Stemming in regional languages using INLTK and Stanza: Not Available